Explanation: Why Two Wager Tiers Exist

This page explains the idea behind Safe Bet and Smart Bet. For the full math, the results and the citations, read `docs/white-paper.md`.

Two different questions

A wager decision can answer two different questions. This project treats them as two tools, not one blended answer.

Question 1: "What wager can't hurt me if I'm right?"

This question ignores confidence. It assumes that each opponent makes their most dangerous legal move: they bet everything and answer correctly. It asks which wager still wins outright. This project calls the answer **Tier 1**. The app shows it as **Safe Bet**.

Question 2: "Given how sure I am, what wager gives me the best real chance to win?"

This question needs two more inputs: your confidence in the category, and a model of how opponents wager and answer. This project calls the answer **Tier 2** for Final Jeopardy and the **Daily Double engine** for a mid-game Daily Double. The app shows both as **Smart Bet**.

How ties are scored

Since the 24 November 2014 episode, a tie for first after Final Jeopardy goes to a tiebreaker clue. Only the winner of that clue returns. Smart Bet gives each tied player an equal chance in the tiebreaker. A two-way tie is half a win. A three-way tie is one third of a win. If every player ends at $0 or less, nobody wins.

Tier 1 does not need this rule. It asks for an outright win.

Why Safe Bet and Smart Bet can differ

Safe Bet plans for the worst case every time. Smart Bet uses your confidence and the likely moves of real contestants.

- With low confidence, Smart Bet can recommend less than Safe Bet. A large wager loses more when you miss, and most opponents do not bet the worst case against you.
- A wager that only ties a rival is worth about half a win. One more dollar turns that tie into a win. For this reason, Smart Bet usually takes the extra dollar when a tie is in play.
- Often a range of wagers gives the same win chance. Then Smart Bet uses a second rule: it picks the wager in the range that wins the most money on average. The winner keeps their final total, so money is the fair tiebreak. Example: you have $12,000, and your rivals have $8,000 and $6,000. Safe

> Bet is $4,001. At 50%, 70% or 90% confidence, every wager from $4,001 to $4,499 gives the same win chance, so Smart Bet recommends $4,499.

Why Smart Bet uses money to break ties

Before version 2.2, Smart Bet took the smallest of the tied wagers. A user found the problem: in a solo game at 100% confidence, Smart Bet said $0. A solo player always wins, so every wager tied, and the smallest was $0. But the player keeps the money. So Smart Bet now ranks wagers by win chance first and money second. The same rule makes a leader who is locked out and sure of the answer bet big.

How Smart Bet models opponents

Smart Bet plays you against **simulated contestants**. The model gives each opponent a role by score: leader, second place, or third place. Each role has a mix of bet types, for example "cover the doubled score of second place" or "bet everything".

The simulated contestants play like contestants from recent seasons (Seasons 38-41, 2021-2025). Play changed in the 2010s: players now jump to high-value clues to hunt Daily Doubles, and they bet them hard. A model of older games does not show this. So the model is calibrated to the modern era:

- The leader almost always bets to cover second place.
- Players who are in front answer Final Jeopardy correctly more often than players behind them.
- Future Daily Doubles can fall at any point in the rest of the round. Leaders find them more often, and finders bet a large part of their score.

Second place's wager mix has no modern source. It still comes from a published study of games from 1990s-2012 (Tesauro et al., 2013). Third place's bets do not depend on the scores, also from that study. `engine/behavior.py` marks each value as CALIBRATED, PUBLISHED or ASSUMED.

The calibration used season statistics from J! Archive. Its terms of use forbid republishing data from it. So no document in this project gives the source statistics or the calibrated values. The values are in `engine/behavior.py` only.

Right and wrong answers are correlated across players. A hard clue is hard for everyone. So when you miss, a rival is more likely to miss too. The model includes you in this correlation.

`docs/adr/0003-simulated-contestant-behavior.md` and `docs/adr/0004-modern-era-calibration.md` give the sources, the decisions and their negative consequences.

Why the Daily Double engine gives a range, not one number

A Final Jeopardy wager ends the game. A Daily Double wager changes the rest of the game: more clues, maybe another Daily Double, then Final Jeopardy. No exact formula covers this. The engine simulates the

rest of the game thousands of times for each candidate wager. It then counts the win share of each wager.

Each simulated result has some uncertainty, as a poll has a margin of error. When the simulation cannot separate some wagers, the app shows a shaded **plateau** on the chart. The app recommends the smallest wager in the plateau. The evidence does not support a larger risk.

One case is different. Some wagers give exactly the same result in every simulated game, for example in a solo game. More simulation cannot separate them. For these exact ties, the engine picks by money: if you are more likely right than wrong, it recommends the largest of them.

What is proven, and what is assumed

The Safe Bet math is fully determined. It needs no assumption about the accuracy of any player.

Smart Bet and the Daily Double engine need a model of opponents. Part of that model is calibrated to modern-era play: Final Jeopardy accuracy by score rank, the outcome correlation, the leader's cover share, Daily Double accuracy, the Daily Double wager size and who finds a Daily Double. Part is published from older games (1990s-2012): the second-place wager mix in one score regime. The other parts are assumptions: the split of the leader's bets, the second-place mix in the other five regimes, the third-place shape, equal tiebreaker odds, and the regular-clue model. A reader cannot check the calibrated values against their source, because the source data is not published. Treat each Smart Bet percentage as a model output, not as a validated statistic. See `RISK_REGISTER.md` (risks #5, #7, #14 and #15).

Related pages

- `docs/white-paper.md` — the math, the results and the citations.
- `docs/tutorial/getting-started.md` — see both tiers in the app.
- `docs/reference/engine-api.md` — the functions behind these ideas.
- `docs/adr/0003-simulated-contestant-behavior.md` — the opponent model.
- `docs/adr/0004-modern-era-calibration.md` — the modern-era calibration.